

No. 190 Recommendation for Vessel Asset Inventory for Computer-based Systems

(June 2025)

1 Introduction

1.1 General

This Recommendation is intended to provide additional information concerning the Vessel Asset Inventory for Computer Based Systems (CBS). Please note that the Vessel Asset Inventory is already required by IACS UR E26 and UR E27. Please see paragraph 4.1.1 of UR E26 and paragraph 3.1.1 of UR E27.

Please see Appendix 1 for a template of the Vessel Asset Inventory and Appendix 2 for a sample of a Vessel Asset Inventory.

2 References

For the purpose of this recommendation, the following IACS Unified Requirements are referenced:

- UR E22 – Computer based systems
- UR E26 – Cyber Resilience of ships
- UR E27 – Cyber Resilience of on-board systems and equipment

3. Rationale for the Vessel Asset Inventory (paragraph 4.1.1.2 of UR E26)

The inventory of CBSs onboard and relevant software used in OT systems is essential for an effective management of cyber resilience of the ship, the main reason being that every CBS becomes a potential point of vulnerability. Cybercriminals can exploit unaccounted and out of date hardware and software to hack systems. Moreover, managing CBS assets enables Companies understand the criticality of each system to ship safety objectives.

4. Requirements for the Vessel Asset Inventory

The full requirements for the Vessel Asset Inventory are in paragraph 4.1.1 of UR E26 and paragraph 3.1.1 of UR E27.

5. Vessel Asset Inventory during the entire life of the ship

UR E26 requires that the Vessel Asset Inventory shall be kept updated during the entire life of the ship. Software and hardware modifications potentially introducing new vulnerabilities or modifying functional dependencies or connections among systems shall be recorded in the inventory.

If confidential information is included in the Vessel Asset Inventory (e.g. IP addresses, protocols, port numbers), special measures shall be adopted to limit access to such information only to authorized people.

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- 5.1 The Vessel Asset Inventory will be used and will need to be kept updated during the entire life of the ship, including:
- Design phase (paragraph 4.1.1.4.1 of UR E26)
 - Construction phase (paragraph 4.1.1.4.2 of UR E26)
 - Commissioning phase (paragraph 4.1.1.4.3 of UR E26)
 - Operation phase (Annual Surveys and Special Surveys) (paragraph 4.1.1.4.4 of UR E26)
- 5.2 The Vessel Asset Inventory will be used by the relevant stakeholders, including designers, system integrators, ship owners, third party equipment and system suppliers, class society design review engineers and class society surveyors.
- 5.3 Since the Vessel Asset Inventory will be used during various times throughout the entire life of the ship, the date or version of the Vessel Asset Inventory should be indicated unless it is managed in a computer-based application, so that any of the various stakeholders will also know if they are looking at the current UpToDate version (and not an out-of-date version).

6. Information in the Vessel Asset Inventory

In order for the Vessel Asset Inventory to meet the intent of effective management of cyber resilience of the ship (mentioned in the rational above), the Vessel Asset Inventory should include beneficial information about the items in the inventory. Please see Appendix 1 for a template of the Vessel Asset Inventory and Appendix 2 for a sample of a Vessel Asset Inventory.

The template is an example of the format for the Vessel Asset Inventory and could be a word document table, excel spreadsheet or similar type file. Other types of files, such as databases or other specialized files, can also be considered, however they should provide the same level of information (or an additional level of information, please refer to paragraph 7 below).

6.1 Columns in the template

For a better understanding of the information that should be indicated in the Vessel Asset Inventory, the following additional information has been provided for each column in the template.

Column A:

Item Number

This column should indicate an item number for each row in the inventory. This is basic information to help identify each equipment item in the inventory.

Column B: (paragraph 4.1.1.3 in UR E26 requirement)

Ship functions and systems

This column should indicate the relevant ship function or system for the subject equipment. For most equipment, this column should indicate one of the items listed below. The list is from paragraph 1.3.2.a of UR E26 which describes the “systems in scope”.

- Propulsion
- Steering
- Anchoring and mooring
- Electrical power generation and distribution
- Fire detection and extinguishing systems

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- Bilge and ballast systems, loading computer
- Watertight integrity and flooding detection
- Lighting (e.g. emergency lighting, low locations, navigation lights, etc.)
- Any required safety system whose disruption or functional impairing may pose risks to ship operations (e.g. emergency shutdown system, cargo safety system, pressure vessel safety system, gas detection system, etc.)
- Navigational systems required by statutory regulations
- Internal and external communication systems required by class rules and statutory regulations

The list from paragraph 1.3.2.b of UR E26 also includes any interface from CBSs in scope of this UR to any other systems. Examples of such cases are, but not limited to, the following:

- passenger or visitor servicing and management systems
- passenger-facing networks
- administrative networks
- crew welfare systems
- any other systems connected to OT systems, either permanently or temporarily (e.g. during maintenance).

In such cases “other” could be entered into this column along with a description. Such as “other – administrative network”.

Column C: (paragraph 4.1.1.3.1 in UR E26 requirement) System

This column should indicate the system name that will help identify the subject equipment. The system name should indicate the most relevant description of the system and should complement the information already provided in Column B. Therefore, Column C does not need to repeat the information from Column B.

In cases when “other” is entered in Column B, Column C should provide more specific information that will help identify the subject equipment.

Column D: (paragraph 4.1.1.3.1 in UR E26 requirement) Equipment

This column should indicate the equipment name that will help identify the subject equipment.

Column E: (paragraph 4.1.1.3.1 in UR E26 requirement) System Category

This column should indicate the System Category defined in UR E22 3.1. Most will be either Category II or Category III. Some will be Category I.

It is possible the “not applicable” could be entered in this column for equipment that does not fit into one of the three system categories defined in UR E22 3.1. One example would be navigation equipment which is out of scope of UR E22. However, navigation equipment is within the scope of UR E26 and therefore should be in the inventory.

Column F: (paragraph 4.1.1.3.1 in UR E26 requirement) Brand/Manufacturer

This column should indicate the brand name or manufacturer name that will help identify the subject equipment.

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Column G: (paragraph 4.1.1.3.1 in UR E26 requirement)
Model/Type

This column should include the model identifier or type identifier that will help identify the subject equipment.

Column H: (paragraphs 4.2.2 and 7.1.1 in UR E22)
Unique identifier

This column should include the system identification information required by paragraphs 4.2.2 and 7.1.1 of UR E22. This is for Category II and Category III equipment. This is not required for Category I, therefore in case the subject equipment is Category I, this column may indicate “not applicable”.

Column I: (paragraph 4.1.1.3.1 in UR E26 requirement)
Operating System - (OS) Version and Patch level included

This column should indicate the OS installed on the subject equipment (this includes physical and virtual machines). The name of the OS should be indicated (such as Linux, Windows, etc.) as well as the version and patch level.

Column J: (paragraph 4.1.1.3.1 in UR E26 requirement)
Firmware -Version and Patch level included

This column should indicate the firmware installed on the subject equipment, as well as the version and patch level. This includes firmware installed on embedded devices and implement the operation of software on hardware, such as RTOS (Freertos, ucos, ...), software installed on switches/routers, etc.

Column K: (paragraph 4.1.1.3.2 in UR E26 requirement)
Software and version

This column should indicate the application software and utility software for the subject equipment, as well as the version. This includes programs that implement functions of onboard systems, and programs that are useful for files managing, system monitoring and protection, process and service management, backups and restore utilities.

Column L: (paragraph 4.2.4.3.1 in UR E26 requirement)
Location Onboard

This column should indicate the location of the subject equipment. It should also indicate the physical access restrictions that are provided (required for Category II and Category III).

Column M: (paragraph 4.1.1.3.1 in UR E26 requirement)
Security zone

This column should identify the security zone that the subject equipment belongs to.

Column N: (paragraph 4.1.1.3.1 in UR E26 requirement)
Short description of functionality/purpose

This column should indicate a short description of the functionality/purpose. This information is needed for hardware and software items.

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Column O: (recommended)

Connections to components and systems within the scope

This column should indicate the connections to components and systems that are within the scope of E26. The intent of this column is to identify the connections that exist with other components and systems that are within the scope of UR E26. The intent of this column is also to determine that all connections have been identified and whether they are within scope or outside scope (Please also see the description for Column P concerning connections to an “Untrusted network”).

Column P: (recommended)

Connections to Untrusted networks

This column should indicate the connections to components and systems that may appear to be not “in the scope of UR E26”. Please note that UR E26 considers any network outside the scope of applicability of UR E26 to be an “Untrusted network”. The intent of this column is to identify possible connections to an untrusted network. In some cases, the subject equipment may not have any connections to an untrusted network. For additional information, please see paragraph 4.2.6 of UR E26 and paragraph 4.2 of UR E27.

Column Q: (paragraph 4.1.1.3.1 in UR E26 requirement)

Physical interfaces

This column should indicate the physical interfaces for the subject equipment (such as network or serial interfaces). The type and number should be indicated, such as USB*2, RJ45*1, serial, etc.

Column R: (paragraph 4.1.1.3.1 in UR E26 requirement)

Supported communication protocols

This column should indicate the relevant supported communication protocols for the subject equipment. This should include general and private protocols, such as HTTP, SSH, NMEA 0183, MODBUS-TCP, etc.

Column S: Recommended (Optional – only if permitted by the Owner’s cyber-security policy)

IP ranges/MAC addresses of nodes connected (if applicable)

This column should indicate the IP ranges and MAC addresses of the nodes connected (if applicable).

Column T: (recommended)

Certificate: Type approval, ship-specific approval, etc. (if applicable)

This column should indicate information concerning any approval by a Class Society, such as a type approval, ship-specific approval, etc. (if applicable). The Class Society and certificate number should be indicated for the system or component that has been issued an approval covering the requirements in UR E26/E27.

Column U: (recommended)

Excluded (If part of an excluded system in accordance with UR E26, Section 6)

This column should remain blank for most equipment. This column should indicate an “X” if the equipment is part of a computer-based system (CBS) that has been excluded from meeting the requirements of UR E26. See Section 6 of UR E26, which would include

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submission of the “Risk assessment for exclusion of CBS from application of requirements” which would be subject to review by the Class Society.

Column V: (paragraph 4.1.1.3 in UR E26 requirement)
Update log

This column should indicate an update log for the subject items. Software and hardware modifications potentially introducing new vulnerabilities or modifying functional dependencies or connections among systems should be indicated in the update log column. Information should indicate when the modification was made, what the modification was and provide direction to Change Records.

6.2 Rows in the template

For a better understanding of the information that should be indicated in the Vessel Asset Inventory, the following additional information has been provided.

Rows may be arranged so that equipment associated with the same system may be grouped together, if this would help the information be displayed in a clear manner.

In some cases, it may be difficult to have all the rows associated with the same system to be grouped together; therefore, grouping is not a requirement. All items are still required to be in the Vessel Asset Inventory.

In some cases, equipment may be installed in between systems (such as a firewall). Accordingly, the boundary firewall can be included in the template in a row associated with the system or can be indicated in a separate row for the individual firewall.

7. Additional level of information for the Vessel Asset Inventory

If any of the stakeholders thinks that information in addition to what is described in paragraph 6 above would be beneficial to be include in the Vessel Asset Inventory, IACS would support the inclusion of the additional level of information. However, the inclusion of the additional level of information should not result in a Vessel Asset Inventory that is more difficult to understand, keeping in mind that there will be various users (please see paragraph 5.2 above).

Appendix 1: A template of the Vessel Asset Inventory

Vessel Asset Inventory		See IACS Recommendation 190 for more information for each column		Vessel Name:			IMO Number:		Class Society Identification Number: (Recommended)		Shipyard Name and Shipyard Identification Number: (Recommended)									Date:	
Column A:	Column B:	Column C:	Column D:	Column E:	Column F:	Column G:	Column H:	Column I:	Column J:	Column K:	Column L:	Column M:	Column N:	Column O: (Recommended)	Column P: (Recommended)	Column Q:	Column R:	Column S: (Recommended)	Column T: (Recommended)	Column U: (Recommended)	Column V:
Item Number	Ship functions and systems	System	Equipment	System Category	Brand/Manufacturer	Model/Type	Unique identifier	Operating System - (OS) Version and Patch level included	Firmware - Version and Patch level included	Software and version	Location	Security zone	Short description of functionality/purpose	Connections to components and systems within the scope	Connections to Untrusted networks	Physical interfaces	Supported communication protocols	IP ranges/MAC addresses of nodes connected (if applicable)	Certificate: Type approval, ship-specific approval, etc. (if applicable)	Excluded (if part of an excluded system in accordance with UR E26, Section 6)	Update log (Check the change records to find details)
1																					
2																					
3																					
4																					
5																					

Appendix 2: A sample of a Vessel Asset Inventory

Vessel Asset Inventory		See IACS Recommendation 190 for more information for each column		Vessel Name:			IMO Number:		Class Society Identification Number: (Recommended)		Shipyard Name and Shipyard Identification Number: (Recommended)									Date:	
Column A:	Column B:	Column C:	Column D:	Column E:	Column F:	Column G:	Column H:	Column I:	Column J:	Column K:	Column L:	Column M:	Column N:	Column O: (Recommended)	Column P: (Recommended)	Column Q:	Column R:	Column S: (Recommended)	Column T: (Recommended)	Column U: (Recommended)	Column V:
Item Number	Ship functions and systems	System	Equipment	System Category	Brand/Manufacturer	Model/Type	Unique identifier	Operating System - (OS) Version and Patch level included	Firmware - Version and Patch level included	Software and version	Location	Security zone	Short description of functionality/purpose	Connections to components and systems within the scope	Connections to Untrusted networks	Physical interfaces	Supported communication protocols	IP ranges/MAC addresses of nodes connected (if applicable)	Certificate: Type approval, ship-specific approval, etc. (if applicable)	Excluded (if part of an excluded system in accordance with Section 6)	Update log (Check the change records to find details)
1	Steering	Steering system	Steering control system	Cat III	SCSUSUK	SCS	ST01abc	Op Sys X v1.007	Industry PLC Corp's PLC firmware v2.023	Steer Straight v03	Wheelhouse	Steering Zone	Steering control system providing control of the rudder	VDR	N/A	RS485 (2/2)	Modbus	IP XXX.XXX	Product-by-Product approval		No update
2	Propulsion	Main Propulsion System	Main Engine Control System (ECS) - Main Unit	Cat III	ABC Machinery	MU-01-LNG	ECS-xx-xx-xx-xA-Bx	Windows 10 22H2	-	MU-01-LNG v1.0	Engine Room	Propulsion Zone	Control system functions and integrate their subsystems.	VDR, RCS, WIAS	Shore	LAN (2/2) USB (0/4) Serial (1/1)	TCP/IP	192.168.xx.x:xxx	24CYxxx		No update
3	Propulsion	Main Propulsion System	Main Engine Control System (ECS) - Wheelhouse Panel	Cat III	XYZ Display	TFT10.4TP	ECS-xx-xx-xx-xA-Bx	Windows 10 22H2	-	TP-WH-LNG v1.0	Wheelhouse	Propulsion Zone	Control panel for main engine which is located in the wheelhouse			LAN (1/2) USB (1/2)	TCP/IP	192.168.xx.x:xxx	24CYxxx		No update

4	Propulsion	Main Propulsion System	Main Engine Control System (ECS) - Control Room Panel	Cat III	XYZ Display	TFT10.4TP	ECS-xx-xx-xx-xA-Bx	Windows 10 22H2	-	TP-CR-LNG v1.10	Control Room	Propulsion Zone	Control panel for main engine which is located in the control room				LAN (1/2) USB (1/2)	TCP/IP	192.168.xx.x:xxx	24CYxxx		2024/08/26 10:05 (UTC) update software to version 1.05 2024/09/26 12:35 (UTC) update software to version 1.10
5	Propulsion	Main Propulsion System	Ethernet Switch	Cat III	DE networks	12m55G	ECS-xx-xx-xx-xA-Bx	NA	DE v1.0	NA	Control Room	Propulsion Zone	Managed ethernet switch which provide external and internal IP connections				LAN (7/12)	TCP/IP	192.168.xx.x:xxx	24CYxxx		No update
6	Propulsion	Main Propulsion System	Firewall	Cat III	XYZ Firewall	Gate 09	ECS-xx-xx-xx-xA-Bx	NA	1214	Gate V1.0	Control Room	Propulsion Zone	Firewall manages ethernet switch, controls and monitors incoming and outgoing network traffic based on predetermined security rules.				LAN (4/5)	TCP/IP	192.168.xx.x:xxx	24CYxxx		No update
7	Propulsion	Main Propulsion System	Main Engine Remote Control System (RCS) - Main Unit	Cat III	RCS Systems	RCS-001A	RCS-01-MU	Windows 10 IoT LTSC 2019	-	RCS-001A v1.0	Control Room	Propulsion Zone	Integrates remote control systems	VDR, ECS, WIAS	-		LAN (2/2) USB (1/4) RS485 (2/2)	TCP/IP, Modbus	192.168.xx.x:xxx	23CYxxx		No update
8	Propulsion	Main Propulsion System	Main Engine Remote Control System (RCS) - Wheelhouse Panel	Cat III	Disp RC AG	DP11EX	RCS-01-WH	Windows 10 IoT LTSC 2019	-	RCS-001B v1.0	Wheelhouse	Propulsion Zone	Bridge control functions				LAN (1/2) USB (1/2)	TCP/IP	192.168.xx.x:xxx	23CYxxx		No update
9	Propulsion	Main Propulsion System	Main Engine Remote Control System (RCS) - Control Room Panel	Cat III	Disp RC AG	DP11EX	RCS-01-CR	Windows 10 IoT LTSC 2019	-	RCS-001C v1.0	Control Room	Propulsion Zone	Engine Control Room control functions				LAN (1/2) USB (1/2)	TCP/IP	192.168.xx.x:xxx	23CYxxx		No update

10	Propulsion	Main Propulsion System	Main Engine Remote Control System (RCS) - Engine Room Panel	Cat III	RCS Systems	RCS-001LC	RCS-01-LC	Windows 10 IoT LTSC 2019	-	RCS-001L v1.0	Engine Room	Propulsion Zone	Emergency control functions			LAN (1/2) USB (1/2)	TCP/IP	192.168.xx.x:xxx	23CYxxx		No update
11	Propulsion	Main Propulsion System	Main Engine Remote Control System (RCS) - Ethernet Switch	Cat III	ESWX	100SW16	RCS-01-SW	NA	ESWX v1.0	NA	Control Room	Propulsion Zone	Connect overall systems and the external networks, additionally manages serial networks			LAN(6/12) RS485 (1/2) RS422 (2/2)	TCP/IP, Modbus	192.168.xx.x:xxx	23CYxxx		No update
12	Watertight integrity and flooding detection	Water level detection and alarm system	Water Ingress Alarm System (WIAS) Alarm Unit	Cat II	SEALEVELSWITCH	SEAAU-B	WIA-AU01	SEA WIA v1.0	Industry PLC Corp's PLC firmware v2.055	SEA Alarm v1.0	Wheelhouse	Propulsion Zone	Integrate information from sensors and provide alarm when they sense any events at the intended high level.	ECS, RCS, VDR		LAN (1/1) RS422 (2/2)	TCP/IP, Modbus	192.168.xx.x:xxx	22CYxxx		No update
13	Watertight integrity and flooding detection	Water level detection and alarm system	Water Ingress Alarm System (WIAS) Sensor	Cat II	SEALEVELSWITCH	SEASW-A	WIA-S01	-	-	-	Cargo Hold	Propulsion Zone	Water level sensor - air purge type			RS422 (1/1)	Modbus	-	-		No update
14	Navigation systems required by statutory regulations	Navigation	Gyro Compass #1	N/A	North Star	NS-333	54293	N/A	-	Gyro XX v1.0	Steering Stand	Navigation zone	Determine north direction and accurate heading	Repeaters, ECDIS	None	USB, RG45, serial	NMEA 0183	IP XXX.XXX	ABS certificate no. DNV certificate no. CCS certificate no. NK certificate no.		No update
15	Navigation systems required by statutory regulations	Navigation	Workstation #1	N/A	Marine Company	NAV-WS-1234	13524	Windows 10 22H2	-	1)Application software v2.0 2)malware portection v1.0 3)backup and restore v1.0 4)Nginx v27.0 5)Vmware v15.0	Wheelhouse Locker #1	Navigation zone	implement of data collection and visualisation, include XX, XX,...(data)	ECDIS	Shore	USB * 3, RJ45 * 2, CD-ROM * 1, RS 232/485/422 *1	MODBUS-TCP(RJ45 2#) NMEA 0183(RS485)	IP XXX.XXX	Ship specific certificate		2024/7/9 09:00 (UTC) update application software to V2.0, check the change records to find details

16	Other - Bounda- ry	Boundar- y (betwee- n systems)	Boundar- y Firewall	Cat III	Firewall PDQ	Bound 101	23412	NA	-	Boundary v1.1	Bridge rack (Locked in rack)	External Connecti- on Zone	ship network access control and malware protection	Navigation System	Shore	USB * 1, RJ45 * 8	All RJ45 support HTTPS, SSH	IP XXX.XXX	25CYxxx		2024/07/ 01 09:00 (UTC) update of virus database 2024/07/ 09 08:00 (UTC) update the software to v1.1
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